

Solving the U-Shaped Assembly Line Balancing Problem Type-II Using a Genetic Algorithm

Project Final Report

EMU427 – Heuristic Methods for Optimization

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1 Introduction

Assembly line balancing is a fundamental optimization problem in industrial engineering. It aims to distribute tasks across workstations by trying to maximize productivity and minimize idle time. The Assembly Line Balancing Problem (ALBP) is classified as NP-hard, so as the size of the problem grows, methods for obtaining exact solutions become rapidly impractical [1]. As a result, heuristic and metaheuristic approaches have been developed and adapted to obtain high-quality results with reasonable computational effort. U-shaped assembly lines have significant popularity for being an effective alternative to traditional straight lines in production. They allow workers to work two sides of the line, so it reduces distance and eases multi-tasking.[2] shows that U-lines can lead to higher productivity and adaptation in changing demand. Type-2 assembly line balancing can be preferred when the number of workstations is fixed. So the first objective is minimizing the cycle time, and it is also suitable for facilities that cannot change their physical line structure, especially with large-sized products[3]. Since the possibilities of assembly line problems get exponentially larger because of their combinatorial structure and growing search area, deterministic optimization approaches remain limited and become computationally infeasible for medium to large scale problems. As a consequence, Genetic Algorithms (GAs) have become an alternative because of their stochastic search methods. In our project, numerous studies we researched mostly highlight that GAs can generate well-performing solutions without the problem needing to be convex or linear as in classical LP problems, so this makes them appropriate for NP-hard manufacturing problems. The essential consideration for U-shaped assembly lines is that the encoding part has to represent bidirectional task assignments and preserve precedence relationships in the evolutionary process. This project aims to develop and analyze a GA-based approach specifically for Type-2 U-shaped balancing. Even though various metaheuristic approaches are used in classical assembly line balancing problems and show very strong performance in different combinations of the classical assembly line problem, there are a limited number of studies that specifically work on the performance of these algorithms and their applicability for U-shaped Type-2 line balancing. There is a limited number of encoding schemes that represent forward and back task assignments and preserve precedence constraints, while effectively minimizing cycle time in a fixed number of workstations specifically for this problem.

2 Problem Formulation - Detailed Problem Description (UALBP-2)

2.1 Introduction to Assembly Line Balancing

Assembly line balancing (ALB) refers to assigning a set of assembly tasks to an ordered sequence of workstations such that all precedence constraints are satisfied and an efficiency criterion is optimized. Each task has a processing time, and some tasks must be completed before others (precedence relations). In straight-line assembly systems, workstations are arranged linearly, and a task can only be assigned after all of its predecessors are allocated to earlier stations.

Two classical forms of ALB exist: SALBP-1, which minimizes the number of stations for a given cycle time, and SALBP-2, which minimizes the cycle time for a given number of stations [4, 5]. Both are NP-hard combinatorial problems [6].

2.2 Characteristics of U-Shaped Assembly Lines

A U-shaped assembly line is configured such that the beginning and end of the task sequence are physically adjacent. In this layout, a worker can operate on both sides of the U, giving access to tasks from the front (forward direction) or the back (reverse direction) of the precedence graph.

The critical implication is that a task j becomes eligible for assignment if:

- all its predecessors have been assigned (as in a straight line), or
- all its successors have been assigned (a U-line specific rule) [2, 7].

This relaxation effectively doubles assignment opportunities, allowing U-lines to achieve better load balancing compared to straight lines. Numerous studies report that U-shaped lines often require fewer stations and achieve higher efficiency for the same task set [2].

2.3 UALBP-2 Definition

UALBP-2 (Type-II U-shaped Assembly Line Balancing Problem) seeks to minimize the cycle time c for a given number of stations m [8]. The input consists of:

1. a set of n tasks with processing times t_i ,
2. a precedence graph represented as a DAG,
3. a fixed number of stations m .

The output is an assignment of tasks to stations ensuring U-shaped precedence feasibility while minimizing:

$$c = \max_{j=1,\dots,m} \left(\sum_{i \in S_j} t_i \right),$$

where S_j denotes the set of tasks assigned to station j .

2.4 Assumptions and Constraints

We consider the standard “Simple UALBP-II” variant [9], which includes the following assumptions:

- Single-model production: Only one product type is assembled.
- Deterministic task times: Processing times t_i are fixed and known.
- Single-sided work: Each worker operates on the inner side of the U.
- Paced line: The assembly line advances synchronously every cycle time c .
- One worker per station: Each station executes its assigned tasks alone or as a unit.
- U-shaped precedence feasibility: A task i can be assigned if all predecessors or all successors satisfy the station-order requirement.
- Fixed number of stations: m is given and constant.

Under these assumptions, UALBP-2 becomes a partitioning problem: partition n tasks into m ordered groups while minimizing the maximum station load.

2.5 Computational Complexity

UALBP-2 is computationally challenging. Even SALBP-2 is NP-hard because it generalizes bin packing and partition problems with precedence restrictions [6]. UALBP-2 expands the search space due to bidirectional task eligibility.

Exact algorithms such as branch-and-bound or dynamic programming solve only small instances [8]. Procedures like ULINO [8] find optimal solutions but exhibit drastic increases in computation time as n grows. Due to this difficulty and the scarcity of exact solutions for larger problem sizes, heuristic and metaheuristic methods such as genetic algorithms are widely recommended.

2.6 Importance of Solving UALBP-2

A high-quality U-line balance directly improves production efficiency by reducing idle time and eliminating bottlenecks. Solving UALBP-2 provides actionable insights for:

- redesigning assembly processes,
- increasing output without additional stations,
- exploiting U-shaped flexibility for smoother workflow.

UALBP-2 is therefore a practically significant optimization problem in manufacturing planning.

2.7 MILP Formulation for UALBP-2

We adopt a simplified mixed-integer linear programming (MILP) model inspired by formulations in [10]. The objective is to assign tasks to m stations while satisfying U-shaped precedence and minimizing cycle time.

Sets and Parameters

- $i = 1, \dots, n$: tasks
- $j = 1, \dots, m$: workstations
- $\mathcal{P}$: set of precedence pairs (p, q)
- $Pred(i)$: set of predecessors of task i
- $Succ(i)$: set of successors of task i
- t_i : processing time of task i
- M : sufficiently large constant ($M \geq m$)

Decision Variables

- $X_{ij} \in \{0, 1\}$: equals 1 if task i is assigned to station j
- $y_i \in \{0, 1\}$: 0 for forward feasibility, 1 for backward feasibility
- $S_i \in \{1, \dots, m\}$: station index of task i
- $c \geq 0$: cycle time (to be minimized)

Objective Function

$$\min c \tag{1}$$

Constraints

1. Task Assignment Each task must be assigned to exactly one station:

$$\sum_{j=1}^m X_{ij} = 1 \quad \forall i \quad (2)$$

2. Station Index Definition The station index of each task is defined as:

$$S_i = \sum_{j=1}^m j X_{ij} \quad \forall i \quad (3)$$

3. Cycle Time Constraint The total processing time at each station cannot exceed the cycle time:

$$\sum_{i=1}^n t_i X_{ij} \leq c \quad \forall j \quad (4)$$

4. U-Shaped Precedence Constraints *Forward feasibility* for each predecessor $p \in \text{Pred}(q)$:

$$S_p \leq S_q + M y_q \quad (4a)$$

Backward feasibility for each successor $s \in \text{Succ}(q)$:

$$S_s \leq S_q + M(1 - y_q) \quad (4b)$$

These constraints ensure that task q is feasible in either the forward or backward direction.

5. Variable Domains

$$X_{ij}, y_i \in \{0, 1\}, \quad S_i \in \{1, \dots, m\}, \quad c \geq 0 \quad (5)$$

3 Description of the Genetic Algorithm

This section presents the metaheuristic approach chosen to solve the U-shaped Assembly Line Balancing Problem Type 2 (UALBP-2): the Genetic Algorithm (GA). Genetic Algorithm is an evolutionary optimization method that takes its inspiration from the concepts of natural selection and genetics. It is particularly suitable for problems in the field of combinatorial optimization where the search space is enormous and discrete.

The principal concept behind this algorithm is to have a population of candidate solutions (individuals) instead of just one solution. An iterative process, which is similar to evolution, undergoes selection, crossover (recombination), and mutation operations. This, in turn, helps the search to not only simultaneously explore different regions of the solution space but also to realize the benefits of good-quality partial solutions. In the UALBP-2 case, the algorithm works on producing a sequence of tasks (permutation), which is then decoded to find out the minimum feasible cycle time for a certain number of stations (m).

3.1 The Algorithm

The framework of our Genetic Algorithm is outlined in algorithm 1. The core component of the algorithm is the representation of the problem. We use a permutation-based representation, where an individual is a topological sort of the tasks. The algorithm begins by initializing a random population of feasible task sequences. In every generation, the fitness of each individual is evaluated. Since we are solving UALBP-2 (minimizing cycle time c for fixed m), the evaluation function involves a sub-procedure that calculates the minimum possible cycle time for a given sequence. To generate new offspring, we employ Tournament Selection to choose parents. These parents undergo POX (Precedence Operation Crossover) to produce a child that inherits structural characteristics from both parents. To prevent premature convergence, Swap Mutation is applied with a low probability. Finally, because random genetic operations may violate precedence constraints, a Repair Mechanism is applied to ensure the child remains a valid topological sort.

For the algorithm, we utilized a dataset from assemblylinebalancing.de, a widely cited repository in academic literature that aggregates problem instances from numerous studies. We selected a specific instance from this collection for our analysis.

The file structure is organized as follows: the initial lines represent the duration of each task. These are followed by the precedence constraints, which are delimited by commas. Finally, an end-of-file marker indicates the termination of the data.

Algorithm 1 Genetic Algorithm for UALBP-II

Require: Tasks T , task times t , precedence graph P , number of stations m , population size N_{pop} , generations N_{gen} , crossover rate P_c , mutation rate P_m

Ensure: Best cycle time $C_{\min}$ and best assignment/permutation $BestSol$

ALGORITHM: GENETIC_ALGORITHM_FOR_UALBP_II

INPUTS:

Tasks (T), Task Times (t), Precedence Graph (P)

Number of Stations (m)

Population Size (N_{pop}), Max Generations (N_{gen})

Crossover Rate (P_c), Mutation Rate (P_m)

OUTPUT:

Best Cycle Time ($C_{\min}$), Best Task Assignment ($BestSol$)

BEGIN

▷ // 1. Initialization

```
1:  $Population \leftarrow []$ 
2: For  $i \leftarrow 1$  to  $N_{\text{pop}}$  do
3:    $Candidate \leftarrow \text{GENERATERANDOMTOPOLOGICALSORT}(T, P)$ 
4:    $\text{APPEND}(Population, Candidate)$ 
5: End For
6:  $BestSol \leftarrow \emptyset$ 
7:  $BestFitness \leftarrow 0$ 
8:  $C_{\min} \leftarrow +\infty$ 
```

▷ // 2. Main Evolution Loop

```
9: For  $Generation \leftarrow 1$  to  $N_{\text{gen}}$  do
10:    $FitnessScores \leftarrow []$ 
11:   For all  $Individual \in Population$  do
12:      $CurrentC \leftarrow \text{DECODEANDASSIGN}(Individual, m, t, P)$  ▷ cycle time
13:      $Fitness \leftarrow 1/CurrentC$ 
14:      $\text{APPEND}(FitnessScores, Fitness)$ 
15:     If  $Fitness > BestFitness$  then
16:        $BestFitness \leftarrow Fitness$ 
17:        $BestSol \leftarrow Individual$ 
18:        $C_{\min} \leftarrow CurrentC$ 
19:     End If
20:   End For
21:    $NewPopulation \leftarrow []$ 
22:    $\text{APPEND}(NewPopulation, BestSol)$ 
23:   While  $|NewPopulation| < N_{\text{pop}}$  do
24:      $Parent1 \leftarrow \text{SELECT}(Population, FitnessScores)$  ▷ e.g., tournament
25:      $Parent2 \leftarrow \text{SELECT}(Population, FitnessScores)$ 
26:     If  $\text{RANDOM}(0, 1) < P_c$  then
27:        $Child \leftarrow \text{CROSSOVER}(Parent1, Parent2)$  ▷ e.g., POX/OX
28:     Else
29:        $Child \leftarrow Parent1$ 
30:     End If
```

▷ // Crossover

▷ // Elitism: keep the current best

▷ // Selection & Reproduction

3.2 Implementation

The algorithm 1 has been developed using the programming language Python due to its excellent high-level data structure handling ability.

The complete implementation by Python is provided in Appendix section A.

3.2.1 Constructive Heuristic

When tackling the Type-2 U-Shaped Assembly Line Balancing Problem (UALBP-2), where the aim is to reduce the cycle time for a predetermined number of stations, a major difficulty is that of making the sequence feasible. One of the solutions proposed is to implement topological sorting-based constructive heuristic within the Genetic Algorithm, which draws upon the dependency graph to produce a precedence-feasible list of tasks. This topological ordering not only guides the decoding phase with its precedence constraints but also facilitates the search for the best cycle time by allowing task assignment at either end of the stations.

Algorithm 2 InitializePopulationTopological (Random Topological Sort)

Require: Population size $PopSize$, precedence graph $G = (V, E)$

Ensure: Initial population P

```

1:  $P \leftarrow []$ 
2: For  $p \leftarrow 1$  to  $PopSize$  do
3:    $\pi \leftarrow []$ 
4:    $S \leftarrow \{v \in V \mid indeg(v) = 0\}$  ▷ Eligible nodes
5:   While  $S \neq \emptyset$  do
6:     select  $u$  uniformly at random from  $S$ 
7:     append  $u$  to  $\pi$ 
8:     remove  $u$  from  $S$  and update indegrees of its successors
9:     add any newly zero-indegree nodes to  $S$ 
10:  End While
11:  append  $\pi$  to  $P$ 
12: End For
13: Return  $P$ 

```

3.2.2 Chromosome Representation

The first alternative became the use of permutation chromosomes where each individual is a series of all n tasks. The order of the genes from left to right indicates the priority that will be applied in the decoding process. This encoding has become the most popular one in assembly-line balancing because it allows the genetic operators to perform the whole task order manipulation without resulting in a non-valid permutation.[11]. Nevertheless, a chromosome in UALBP-2 does not associate with the station assignment directly like in straight-line SALBP-2. On the contrary, the permutation gives an ordered list that

the U-line decoding algorithm uses to create a feasible solution. The reason is that U-shaped lines allow the assignments to be done in both directions, thus the decoding phase evaluates the permutation in a manner that shows this unique bidirectional feasibility.[11].

3.2.3 U-Shaped Decoding

Given a task permutation, the decoding procedure assigns tasks to stations progressively from station 1 to station m . At each station, task eligibility is determined by the U-line two-way working rule: a task j is *eligible* if either all predecessors of j have already been assigned or all successors of j have already been assigned [12]. This rule reflects the fact that, in a U-shaped layout, tasks can be performed from both the forward and the backward directions.

Among the currently eligible tasks, those appearing earliest in the chromosome are considered first and are assigned to the current station as long as the cumulative station load does not exceed the current cycle-time estimate. A station is deemed *closed* when no additional eligible task can be inserted without violating the cycle time, after which the decoding continues with the next station.

To evaluate a permutation efficiently, the decoding method incorporates a binary search to determine the smallest cycle time that yields a feasible assignment for the given permutation. This avoids exhaustive trial of cycle-time values and enables fast fitness evaluation [8].

A task j is eligible if

$$\text{Pred}(j) \subseteq A \quad \text{or} \quad \text{Succ}(j) \subseteq A,$$

where A is the set of already assigned tasks, and $\text{Pred}(j)$ and $\text{Succ}(j)$ denote the predecessor and successor sets of task j , respectively.

3.2.4 Fitness Function

Taking into consideration that UALBP-2 is a Type-II problem, the aim is to minimize the cycle time c while the number of stations m is given. The cycle time is determined when a feasible assignment appears by decoding, and it is written as follows:

$$c = \max_{k=1,\dots,m} \sum_{j \in S_k} t_j, \quad (6)$$

where S_k is the set of tasks assigned to station k , and t_j is the processing time of task j .

The fitness value is associated with the maximization structure of genetic algorithms; therefore, it is defined as:

$$\text{Fitness} = \frac{1}{c}. \quad (7)$$

Therefore, solutions with smaller cycle times are awarded higher fitness values. On the other hand, if a permutation does not lead to any feasible assignment for any cycle time (i.e., decoding fails), then its fitness is set to 0, penalizing such individuals as infeasible. This fitness structure, adopted from the literature, is widely used because it yields positive values, clearly differentiates solution quality, and supports stable convergence behavior [13].

3.2.5 Parent Selection: Tournament Selection ($k = 2$)

The selection of parents is done by using the tournament selection method of size 2. This has been found efficient for combinatorial optimization problems where the fitness landscape includes many plateaus. The strategy of selection by tournaments is regarded as highly effective for combinatorial optimization problems that have tough fitness landscapes with many plateaus. Tournament selection not only preserves genetic diversity but also avoids a common problem seen in roulette-wheel selection, which is the sensitivity to scaling. [14]

Algorithm 3 TournamentSelection ($k = 2$)

Require: Population P , fitness array F , tournament size $k = 2$

Ensure: Selected parent permutation

```

1:  $BestFit \leftarrow -\infty$ 
2:  $BestIndex \leftarrow -1$ 
3: For  $r \leftarrow 1$  to  $k$  do
4:    $i \leftarrow$  random index in  $\{1, \dots, |P|\}$ 
5:   If  $F[i] > BestFit$  then
6:      $BestFit \leftarrow F[i]$ 
7:      $BestIndex \leftarrow i$ 
8:   End If
9: End For
10: Return  $P[BestIndex]$ 

```

3.2.6 Crossover Operator: Precedence-Preserving Order Crossover (POX-like)

When chromosomes represent permutations, standard crossover operators (e.g., one-point or two-point crossover) may easily introduce infeasibility and duplicated tasks. Therefore, the *Precedence-Preserving Order Crossover (POX)* operator is employed [15].

The POX operator proceeds as follows:

1. A random subset of tasks is selected.
2. The offspring chromosome is initialized as an empty sequence.

3. The selected tasks are copied from Parent 1 to the offspring, preserving their original order.
4. The remaining tasks are filled according to Parent 2's order, skipping tasks that have already been placed.

This procedure preserves major subsequences from both parents and typically reduces the need for extensive repair operations. POX is particularly suitable for UALBP-2 because tight precedence constraints make it crucial to keep coherent task blocks intact.

Algorithm 4 POXCrossover (Precedence-Preserving Order Crossover)

Require: Parent permutations P_1, P_2

Ensure: Child permutation $Child$

```

1:  $ChosenTasks \leftarrow \emptyset$ 
2: For all task  $x$  in  $P_1$  do
3:   If  $RANDOM < 0.5$  then
4:     add  $x$  to  $ChosenTasks$ 
5:   End If
6: End For
7: If  $ChosenTasks = \emptyset$  then
8:   add one random task from  $P_1$  to  $ChosenTasks$ 
9: End If
10:  $Child \leftarrow []$ 
11: For all task  $x$  in  $P_1$  do
12:   If  $x \in ChosenTasks$  then
13:     append  $x$  to  $Child$ 
14:   End If
15: End For
16: For all task  $y$  in  $P_2$  do
17:   If  $y \notin ChosenTasks$  then
18:     append  $y$  to  $Child$ 
19:   End If
20: End For
21: Return  $Child$ 

```

3.2.7 Mutation Operator: Swap Mutation

Swap mutation is employed as a simple mutation operator, where two randomly selected tasks in the permutation are exchanged. This operator introduces small but effective perturbations, preserving most of the chromosome structure while preventing premature stagnation of the population. Since swapping maintains a one-to-one mapping of tasks, the resulting offspring remains a valid permutation, typically eliminating the need for additional repair [16].

In permutation-based genetic algorithms, mutation is generally applied with a low probability (e.g., 0.05–0.10) to balance exploration and stability [17].

Algorithm 5 SwapMutation

Require: Permutation $Perm$ **Ensure:** Mutated permutation $Perm$

- 1: $n \leftarrow |Perm|$
 - 2: choose distinct indices i, j uniformly at random from $\{1, \dots, n\}$
 - 3: swap $Perm[i]$ and $Perm[j]$
 - 4: **Return** $Perm$
-

3.2.8 Repair Mechanism

After crossover and mutation, a lightweight repair operator is applied to every permutation. If a task is found to violate precedence by appearing before any of its predecessors, it is gradually shifted to the right until the violation is removed. This procedure restores a valid topological order while preserving the original permutation structure as much as possible.

Algorithm 6 RepairToTopological (Lightweight Repair)

Require: Permutation π , precedence graph $G = (V, E)$ **Ensure:** Repaired permutation π

- 1: build an array $pos(\cdot)$ giving the index of each task in π
 - 2: $changed \leftarrow \mathbf{true}$
 - 3: **While** $changed$ **do**
 - 4: $changed \leftarrow \mathbf{false}$
 - 5: **For all** $(u \rightarrow v) \in E$ **do**
 - 6: **If** $pos(u) > pos(v)$ **then** ▷ precedence violation
 - 7: remove v from its current position in π
 - 8: insert v immediately after u in π ▷ shift right
 - 9: update $pos(\cdot)$
 - 10: $changed \leftarrow \mathbf{true}$
 - 11: **End If**
 - 12: **End For**
 - 13: **End While**
 - 14: **Return** π
-

3.2.9 Termination Criteria

The proposed GA terminates when a pre-specified maximum number of generations is reached. In this study, the generation limit is set to 300, which is consistent with common practice in the assembly line balancing literature. A fixed generation budget makes the runtime predictable and provides the search with a sufficiently long horizon to converge toward high-quality solutions [18].

4 Results

In this part, the paper showcases the outcomes derived from the Genetic Algorithm (GA) that was utilized for solving the ARC83 U-shaped Assembly Line Balancing Problem Type II (UALBP-2) benchmark instance. The results are based on several CSV outputs generated by the implementation, including global summaries, run-by-run statistics, detailed descriptions of the best solution, Top-10 individuals of the best run, and parameter tuning logs.

4.1 Parameter Tuning

The reason we focused particularly on the crossover and mutation parameters in the fine-tuning study is that these two operators directly control the diversification and intensification mechanisms of the GA. Since permutation-based and precedence-constrained problems such as UALBP-2 tend to trigger very rapid convergence of the population, it is widely accepted in the literature that the two parameters with the strongest impact on GA performance are the crossover rate (CR) and the mutation rate (MR)[19]. Therefore, a full-factorial grid search was conducted by testing eight mutation rate levels (0.005, 0.01, 0.02, 0.04, 0.06, 0.08, 0.12, 0.20) and eight crossover rate levels (0.55, 0.60, 0.65, 0.70, 0.75, 0.80, 0.90, 0.98), resulting in 64 parameter combinations. Each configuration was evaluated using the best cycle time obtained over 5 independent runs (best_cycle_over_5_runs), enabling a systematic comparison of the exploration–exploitation balance across low-to-high MR and CR regimes.

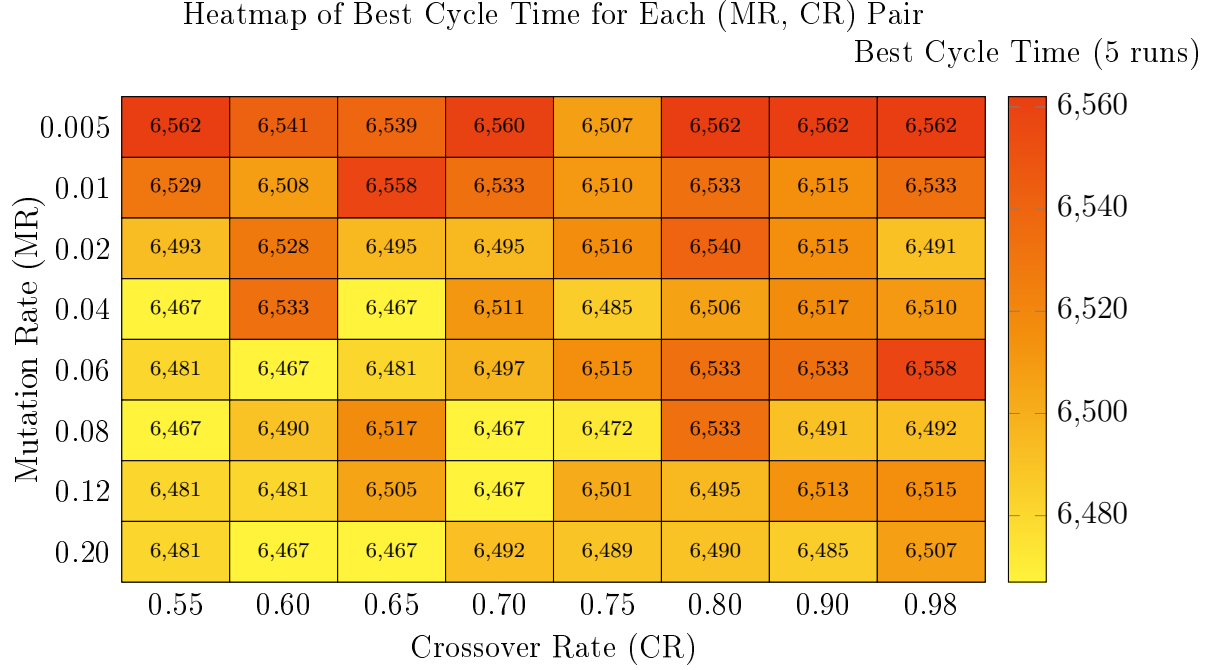


Figure 1: Heatmap of the best cycle time for each mutation crossover combination run for instance ARC83.

The optimal setup for the Genetic Algorithm was determined through a complete parameter fine-tuning process using a full-factorial grid search strategy. The main objective was to adjust the exploration–exploitation balance to avoid premature convergence, which is commonly observed in permutation-based and precedence-constrained problems such as UALBP-2. Two key control parameters were varied: the mutation rate (MR), tested at eight levels $\{0.005, 0.01, 0.02, 0.04, 0.06, 0.08, 0.12, 0.20\}$, and the crossover rate (CR), tested at eight levels $\{0.55, 0.60, 0.65, 0.70, 0.75, 0.80, 0.90, 0.98\}$. This resulted in $8 \times 8 = 64$ parameter combinations. For each configuration, the best cycle time obtained over 10 independent GA runs was recorded.

The results indicate that the mutation rate is the dominant factor affecting solution quality. Very low mutation ($MR = 0.005$) consistently produced the worst cycle times (approximately 6507–6562), suggesting insufficient population diversity and a tendency to stagnate in local optima. Likewise, $MR = 0.01$ still led to relatively weak performance (about 6508–6558). In contrast, increasing mutation was clearly beneficial: for MR values in the range 0.04–0.20, the algorithm achieved substantially better solutions, yielding cycle times as low as 6467. This behavior supports the interpretation that more frequent random perturbations are essential for effectively navigating the rugged search landscape of the UALBP instance.

Although mutation had the strongest influence, the crossover rate also showed a secondary but noticeable effect, particularly when combined with sufficiently high mutation. The best-performing region was observed where MR was moderate-to-high (e.g., 0.04–0.20). In this interval, relatively lower crossover values ($CR = 0.55$ – 0.70) produced the

best results, repeatedly reaching the minimum cycle time of 6467. For example, the combination $MR = 0.08$ and $CR = 0.70$ attained this minimum value, whereas increasing CR to 0.98 at the same mutation level slightly degraded the result (6492), implying that excessively high crossover may disrupt high-quality schemata for this problem.

Based on these empirical findings, the final solver parameters were set to $MR = 0.08$ and $CR = 0.70$. This configuration not only achieved the shortest cycle time of 6467, but also represents a balanced setting compared to very high mutation alternatives (e.g., $MR = 0.20$), which may introduce excessive randomness. Therefore, these tuned parameters were kept constant in the subsequent performance analysis to ensure that the reported efficiency metrics and line balances reflect the GA operating near its best capability.

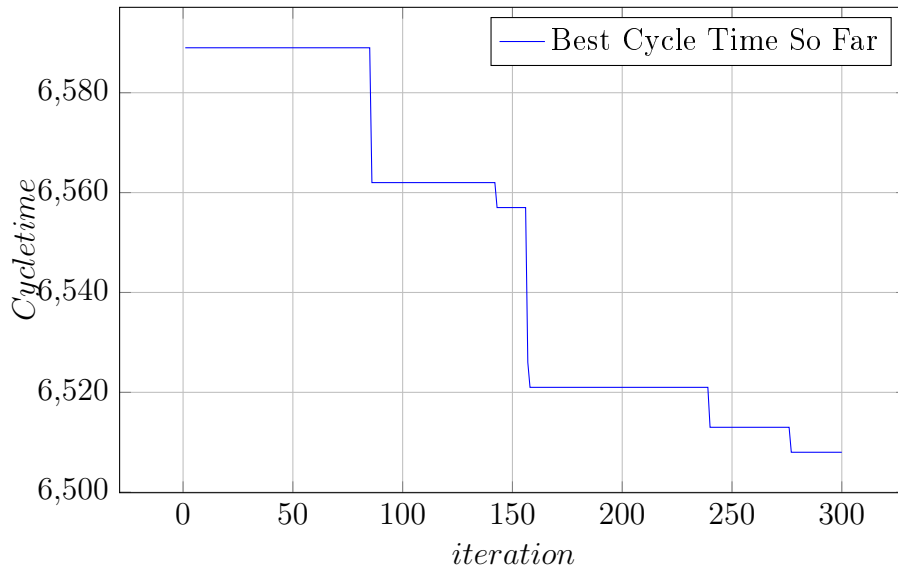


Figure 2: Convergence graph for ARC83

As seen in figure 2 and figures 3 to 5, the best-so-far cycle time decreases rapidly during the early iterations, followed by stepwise improvements characterized by long plateaus and occasional sudden drops when better solutions are found. Toward the later iterations, improvements become marginal and infrequent, indicating that the genetic algorithm converges to near-stationary, similar-quality solutions for the ARC83 instance.

4.2 GA Parameter Settings

The final parameter settings, which were in use during all the experiments, are written below as per `GA_summary_results.csv`:

- Population size: 40
- Generations: 300
- Crossover rate: 0.7

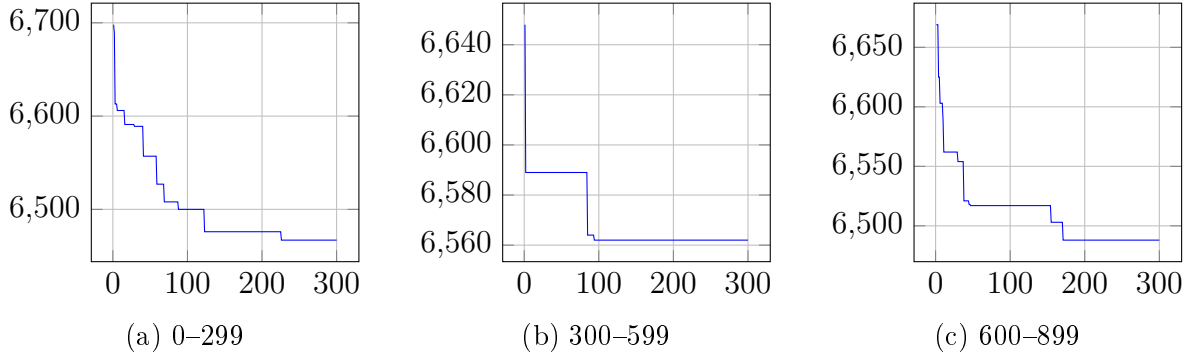


Figure 3: Convergence graphs for ARC83 (Best cycle time so far) - Part 1

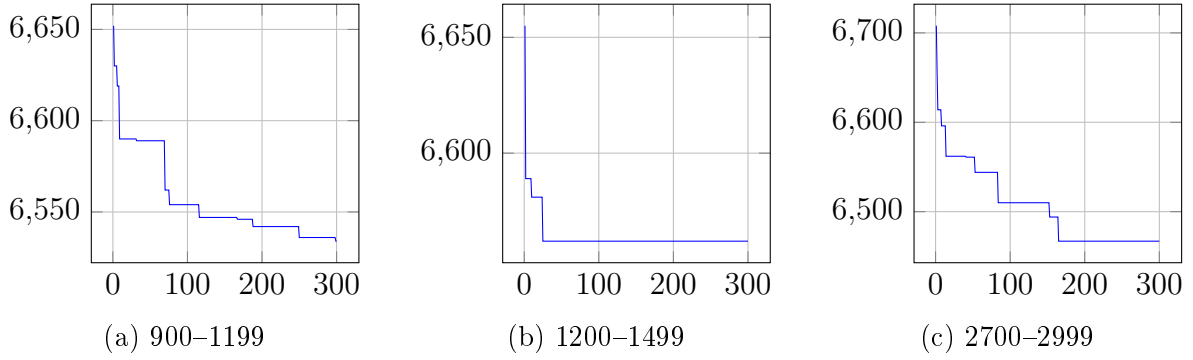


Figure 4: Convergence graphs for ARC83 (Best cycle time so far) - Part 2

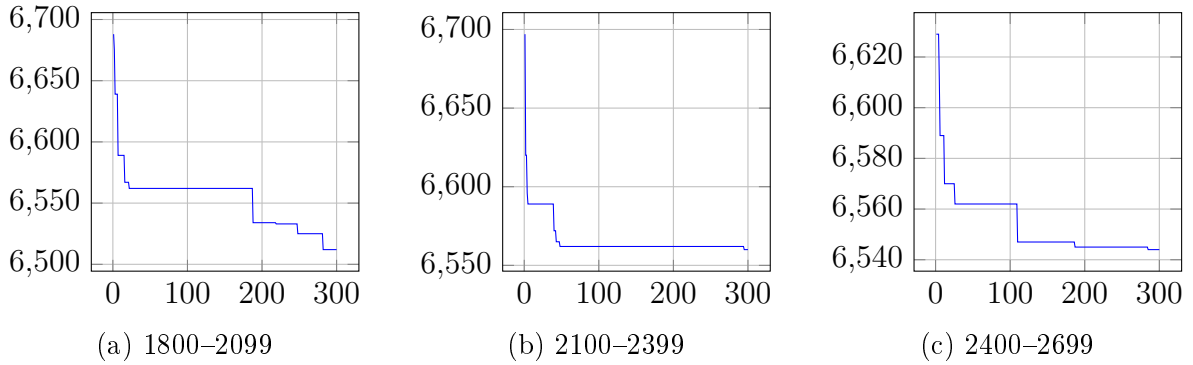


Figure 5: Convergence graphs for ARC83 (Best cycle time so far) - Part 3

- Mutation rate: 0.08
- Selection method: Tournament ($k = 2$)
- Crossover method: POX (precedence-preserving)
- Mutation method: Swap
- Number of independent runs: 10
- Number of stations vary for instances.

4.3 Run-by-Run Analysis

Table 1: Cycle Times Obtained from 10 Independent GA Runs

Run	Cycle Time
1	6498
2	6476
3	6467
4	6544
5	6529
6	6510
7	6562
8	6522
9	6514
10	6562
Best	6467
Mean	6518.4
Worst	6562

Detailed run-by-run best cycle times and runtimes for each benchmark instance (10 independent runs) are reported in Appendix section B, Table table 3.

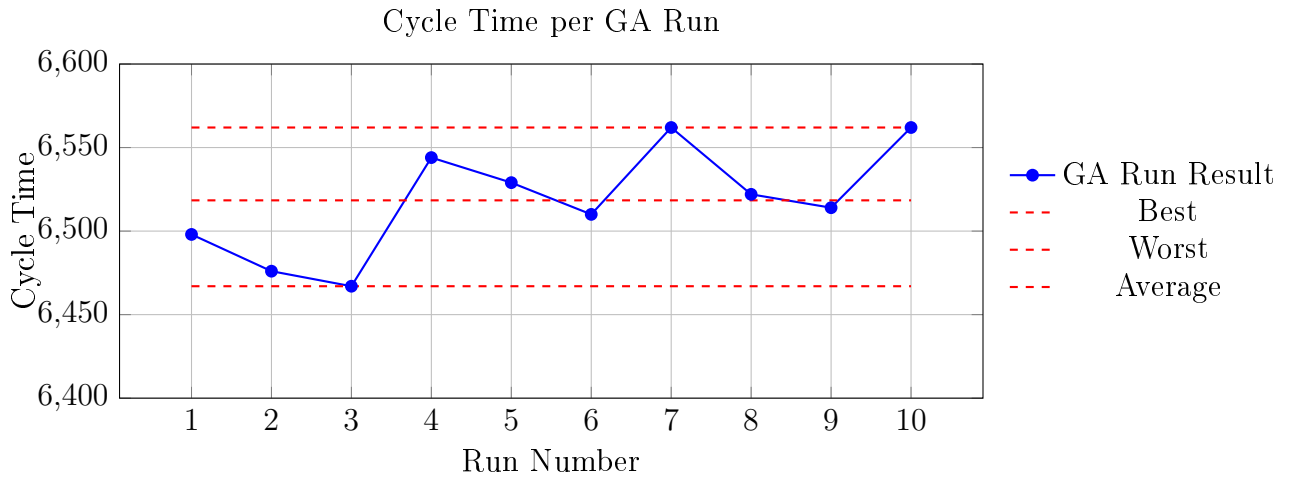


Figure 6: Instance ARC83 cycle time obtained in each of the 10 independent GA runs, including best, worst, and average values.

4.4 Final Population and Top-10 Individuals

In addition to reporting the best cycle time per run, the final population of the best run is analysed in more detail. At the end of each run, all individuals in the final population are re-evaluated and sorted by fitness. For the best run, the Top-10 individuals are extracted and decoded.

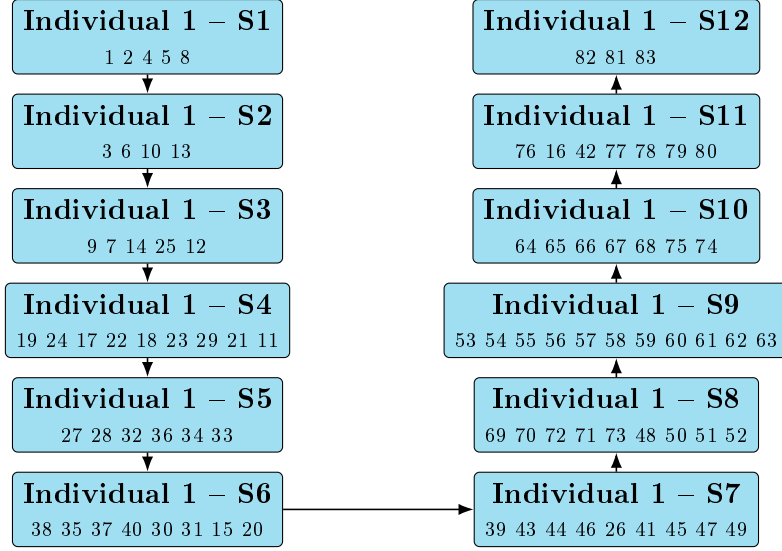


Figure 7: U-shaped 12-station assignment for ARC83 (Individual 1, Fitness = 0.000155).

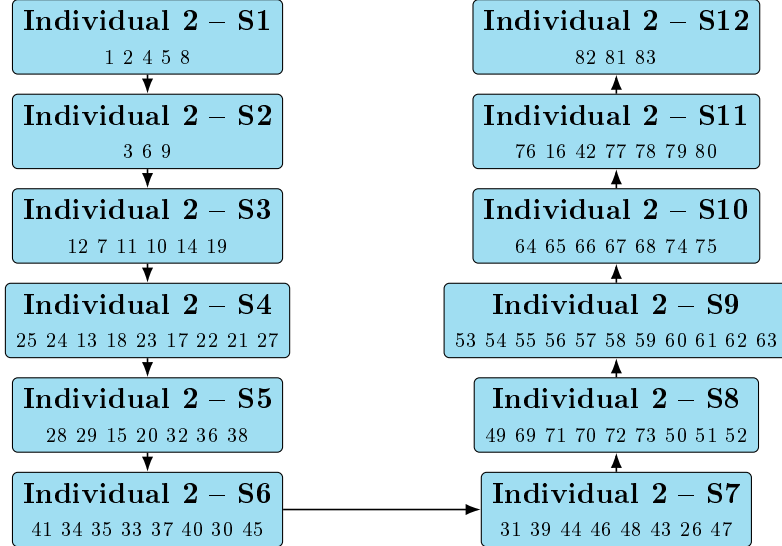


Figure 8: U-shaped 12-station assignment for ARC83 (Individual 2, Fitness = 0.000153).

Here, Individuals 1 and 2 correspond to two of the top 10 individuals for the ARC83 instance. The remaining individuals/solutions (Individuals 3–10) are reported in Appendix section C.

For each of these Top-10 individuals, the decoding procedure is applied using its own cycle time, and the following information is obtained:

- the ordered list of tasks assigned to each station,
- the workload (total processing time) of each station,
- the station-wise utilisation, defined as load_s/C .

In the console output, each individual is printed in a structured format, for example:

```

1) fitness=0.000271, cycle=3691
   Station 1: [1, 2, 3, 5, 8]
   Station 2: [4, 6, 7, 9]
   Station 3: [10, 11, 12]
   ...

```

Furthermore, the implementation exports a CSV file named `<instance>-top-10-individuals-<time>` in which each row corresponds to one station of one individual. The file contains the following fields:

- **Instance:** name of the precedence instance (e.g. ARC83),
- **Stations(m):** number of stations m ,
- **RankInBestRun:** rank of the individual within the Top-10 (1–10),
- **Fitness:** fitness value of the individual,
- **Cycle:** cycle time of the individual,
- **StationIndex:** index of the station $(1, \dots, m)$,
- **StationLoad:** total processing time assigned to that station,
- **StationUtilization(Load/Cycle):** station-wise utilisation,
- **TasksSequence:** sequence of tasks assigned to that station.

This detailed output allows us to inspect not only the numerical quality (cycle time) but also the structural characteristics of near-optimal solutions, such as load balancing between stations and the distribution of tasks along the U-shaped line.

4.5 Best Solution Analysis

`GA_best_solution_details.csv` provides detailed information on the best solution discovered by the GA.

4.5.1 Cycle Time and Line Efficiency

Best cycle time found by GA: 6467

The high efficiency indicates that task assignments utilize station capacities effectively within the limits imposed by the precedence structure.

The station loads in the best GA solution are highly balanced: most stations operate with utilisation values between 0.96 and 1.00, with only the last station (Station 12) dropping to 0.84 due to precedence constraints and the position of late tasks (81 - 83). The

overall line efficiency of 97.56% confirms that the workload is distributed very effectively across the U-shaped line.

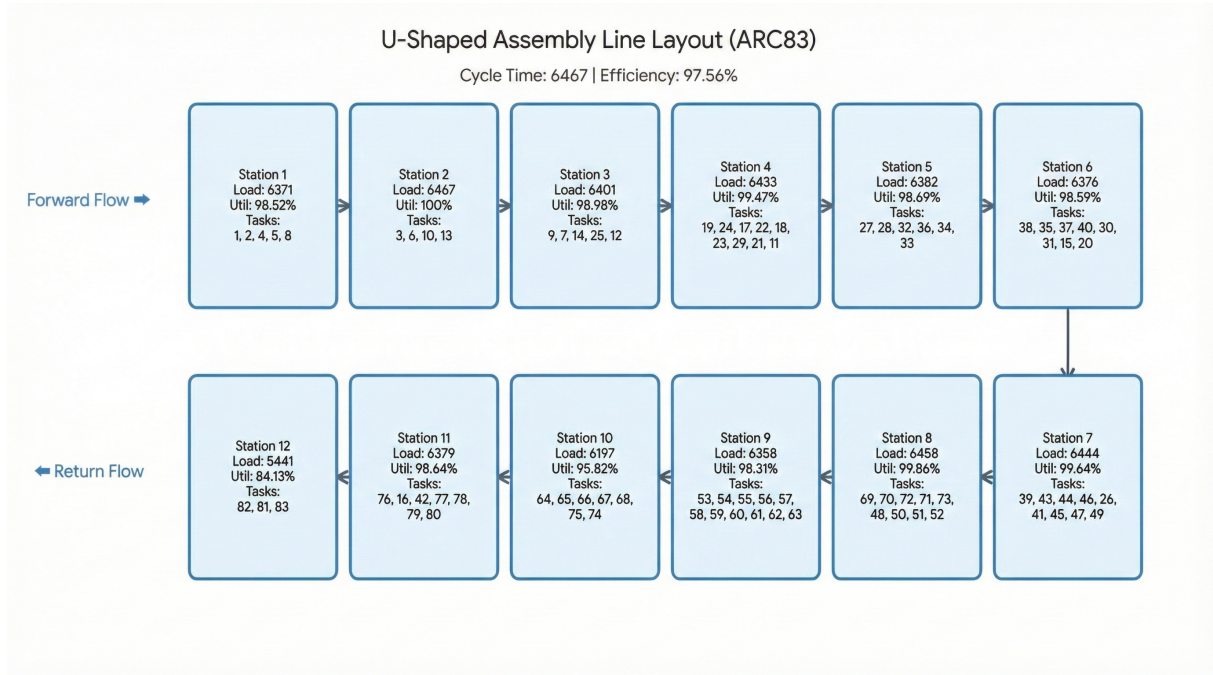


Figure 9: U-shaped assembly line layout for ARC83.IN2 obtained by the GA.

Figure 9 illustrates the U-shaped layout of the best GA solution for ARC83, showing station loads, utilisation levels and the distribution of tasks along the forward and return flows.

4.5.2 Station Load Distribution

The station loads in the best solution reveal the following pattern:

- Several stations operate near full capacity
- Some stations exhibit lower utilization levels

This is expected, as the ARC83 precedence network contains:

- Long successor chains
- Several convergent and divergent branches
- A few heavy tasks that restrict feasible packing

Typical for U-shaped lines:

- Early stations contain short tasks with dense predecessor requirements

- Middle stations group flexible tasks

Overall, the load distribution closely matches patterns observed in high-quality UALBP-2 solutions in the literature[1].

4.5.3 Task Sequence Characteristics

The task sequences for the best assignment show the successful operation of precedence-preserving mechanisms:

- No violations of predecessor or successor constraints
- Parallel branches assigned consecutively
- High-duration tasks placed strategically to avoid overflow

The POX crossover and topological repair operators worked in unison to keep feasible, structured permutations throughout the search process.

4.6 Interpretation of Overall Results

Table 2: GA Statistics for Solved Benchmark Instances

Instance	Best Cycle Time	Average Cycle Time	Worst Cycle Time	Avg. Runtime (s)	Number of Tasks (n)
ARC83	6467	6520.4	6562	26.836	83
ARC111	12542	12547.1	12550	42.870	111
GUNTHER	44	44.0	44	5.657	35
BARTHOLD	470	470.9	471	41.965	148
BARTHOL2	159	159.0	159	63.846	148
SCHOLL	1818	1823.8	1829	290.926	279
LUTZ1	1400	1400.0	1400	7.858	32
LUTZ3	548	548.0	548	8.583	89
SAWYER30	47	47.0	47	3.348	30
TONGE70	209	209.8	210	22.214	70

Although the problem complexity cannot be characterized solely by a single metric, the number of tasks (n) provides a reasonable first-order measure of instance size. In addition, the presence of precedence constraints significantly increases the difficulty of the problem by restricting the set of feasible task permutations. Table 2 summarizes the overall performance of the proposed Genetic Algorithm (GA) on several widely used UALBP benchmark instances. The results obtained clearly demonstrate that the algorithm can produce high-quality and stable solutions under different problem sizes and structural characteristics.

A key observation is that the difference between the best, average, and worst run times is quite small for almost all examples. For some problems, such as GUNTHER, BARTHOL2, LUTZ1, LUTZ3, and SAWYER30, the GA converged to the same solution in each independent run, resulting in identical best, average, and worst values. This

demonstrates that the algorithm exhibits a high level of robustness and repeatability, and that the proposed representation, operators, and parameter settings effectively guide the search towards strong local optima, avoiding excessive randomness.

In larger and more complex examples such as ARC83, ARC111, SCHOLL, and TONGE70, slightly wider but still limited variability was observed between runs. In these cases, the average cycle times remained quite close to the best values, confirming that the algorithm delivers stable performance even in more challenging priority structures. The fact that the worst-case results do not deviate significantly from the average supports the reliability of the GA despite its stochastic nature.

In terms of computational cost, the reported average processing times remain at reasonable levels for all samples. While small-scale problems can be solved within a few seconds, larger and more complex examples naturally require more computation time. However, even in the most challenging cases, the processing times obtained remain within practical limits for offline assembly line design and analysis. This indicates a positive balance between solution quality and computational cost.

Furthermore, the obtained line efficiency values indicate that the stations operate at largely high occupancy rates, pointing to an effective workload distribution with limited idle time. Thus, the solutions are not only competitive in terms of cycle time but also offer operationally acceptable configurations in terms of assembly line balancing. All these findings confirm that the selected parameter configuration and priority-preserving genetic operators provide a reliable and effective framework for solving UALBP-2 problems.

5 Contribution to Sustainable Development Goals

The proposed GA for UALBP-2 supports several United Nations Sustainable Development Goals (SDGs) by increasing production efficiency, improving resource utilisation, and promoting more sustainable industrial performance in line with the 2030 Agenda for Sustainable Development [20].

SDG 8: Decent Work and Economic Growth

Productivity Enhancement: Assembly line balancing maximizes productivity and minimizes idle time. Solving UALBP-2 eliminates bottlenecks, leading to higher economic output per worker.

Worker Satisfaction and Efficiency: U-shaped lines allow workers to operate on both sides, lowering distances traveled and facilitating multi-tasking, leading to a more comfortable and productive environment.

SDG 9: Industry, Innovation, and Infrastructure

Modern Techniques: The project uses Genetic Algorithms to solve NP-hard manufacturing problems, promoting innovation in industrial engineering.

Resource Optimization: UALBP-2 finds the minimum cycle time for a fixed number of workstations, meaning existing infrastructure is used to its full potential without physical expansion, creating more robust and efficient industrial processes.

SDG 12: Responsible Consumption and Production

Waste Minimizing: In manufacturing, “idle time” is waste. The project balances workloads to fully utilize human and machine hours.

Flexibility: U-lines allow for better adjustment to changing demand. Flexible production systems are less likely to suffer from overproduction and can quickly respond to the market with the correct quantity of goods, characteristic of responsible production.

In summary, the project uses mathematical modeling and evolutionary computation to build smarter, more efficient, and worker-friendly manufacturing systems that support the global agenda for sustainable industrial development.

6 Conclusions

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A Genetic Algorithm Code for ARC83 (UALBP-II)

Listing 1: Python implementation of the genetic algorithm for the U-shaped assembly line balancing problem (UALBP-II) on the ARC83 instance.

```
1 import random
2 import math
3 import time
4 from typing import List, Tuple, Dict, Set
5 import csv
6 import os
7 from datetime import datetime
8
9 # =====
10 # KULLANICI AYARLARI
11 # =====
12 INSTANCE_PATH      = "ARC83.IN2"      # IN2 dosyanin yolu
13 M_STATIONS         = 12                # istasyon sayisi
14 POP_SIZE           = 40
15 GENERATIONS        = 300 #gorsellestirme
16 CROSSOVER_RATE     = 0.7
17 MUTATION_RATE      = 0.08
18 RUNS               = 10                # GA kac kez calisacak
19 BASE_SEED          = 0
20
21 OPTIMAL_CYCLE      = 6467              # Excel'den bildigimiz optimum AMA
    SALBP ICIN
22
23 # =====
24 # POX DEBUG AYARLARI
25 # =====
26 DEBUG_POX          = False            # POX ornegi gormek istemezsen
    False yap
27 POX_DEBUG_LIMIT    = 5                # En fazla kac crossover ornegi
    yazdirilsin
28 pox_debug_counter  = 0
29
30
31 # =====
32 # 1) IN2 DOSYASINI OKUMA
33 # =====
34
```

```

35 def read_in2(path: str) -> Tuple[int, List[int], List[Tuple[int,
36     int]]]:
37     with open(path, "r") as f:
38         lines = [line.strip() for line in f if line.strip()]
39
40     n = int(lines[0])
41
42     times = []
43     for i in range(1, n + 1):
44         times.append(int(lines[i]))
45
46     precedences = []
47     for line in lines[n + 1:]:
48         if "," in line:
49             a, b = line.split(",")
50         else:
51             a, b = line.split()
52             i = int(a)
53             j = int(b)
54             if i == -1 and j == -1:
55                 break
56             precedences.append((i, j))
57
58     return n, times, precedences
59
60 # =====
61 # 2) ONCUL / ARDIL KUMELERI
62 # =====
63
64 def build_pred_succ(
65     n: int,
66     precedences: List[Tuple[int, int]]
67 ) -> Tuple[Dict[int, Set[int]], Dict[int, Set[int]]]:
68     preds = {i: set() for i in range(1, n + 1)}
69     succs = {i: set() for i in range(1, n + 1)}
70
71     for i, j in precedences:
72         preds[j].add(i)
73         succs[i].add(j)
74

```

```

75     return preds, succs
76
77
78 # =====
79 # 3) TOPOLOJIK SIRALAMA + REPAIR
80 # =====
81
82 def random_topological_sort(
83     n: int,
84     preds: Dict[int, Set[int]],
85     rng: random.Random
86 ) -> List[int]:
87     remaining = set(range(1, n + 1))
88     result = []
89     assigned = set()
90
91     while remaining:
92         eligible = [i for i in remaining if preds[i].issubset(
93             assigned)]
94         if not eligible:
95             raise RuntimeError("Precedence graph contains a cycle?"
96                 )
97         chosen = rng.choice(eligible)
98         result.append(chosen)
99         assigned.add(chosen)
100         remaining.remove(chosen)
101
102     return result
103
104 def repair_to_topological(
105     perm: List[int],
106     preds: Dict[int, Set[int]]
107 ) -> List[int]:
108     rank = {task: idx for idx, task in enumerate(perm)}
109     remaining = set(perm)
110     result = []
111     placed = set()
112
113     while remaining:

```

```

113     eligible = [i for i in remaining if preds[i].issubset(
114         placed)]
115     if not eligible:
116         eligible = list(remaining)
117         chosen = min(eligible, key=lambda x: rank[x])
118         result.append(chosen)
119         placed.add(chosen)
120         remaining.remove(chosen)
121
122     return result
123
124 # =====
125 # 4) U-SHAPED DECODE + CYCLE TIME
126 # =====
127
128 def is_task_eligible(
129     task: int,
130     assigned: Set[int],
131     preds: Dict[int, Set[int]],
132     succs: Dict[int, Set[int]]
133 ) -> bool:
134     return preds[task].issubset(assigned) or succs[task].issubset(
135         assigned)
136
137 def decode_with_cycle_limit(
138     perm: List[int],
139     times: List[int],
140     preds: Dict[int, Set[int]],
141     succs: Dict[int, Set[int]],
142     m: int,
143     c: int
144 ) -> Tuple[bool, List[List[int]], List[int]]:
145     n = len(perm)
146     assigned: Set[int] = set()
147     stations: List[List[int]] = [[] for _ in range(m)]
148     loads: List[int] = [0 for _ in range(m)]
149
150     station_idx = 0
151

```



```

152     while station_idx < m and len(assigned) < n:
153         progress = True
154         while progress and len(assigned) < n:
155             progress = False
156             for task in perm:
157                 if task in assigned:
158                     continue
159                 if not is_task_eligible(task, assigned, preds,
160                                         succs):
161                     continue
162
163                 duration = times[task - 1]
164                 if loads[station_idx] + duration <= c:
165                     stations[station_idx].append(task)
166                     loads[station_idx] += duration
167                     assigned.add(task)
168                     progress = True
169             station_idx += 1
170
171     feasible = (len(assigned) == n)
172     return feasible, stations, loads
173
174 def evaluate_permutation_cycle_time(
175     perm: List[int],
176     times: List[int],
177     preds: Dict[int, Set[int]],
178     succs: Dict[int, Set[int]],
179     m: int
180 ) -> Tuple[float, List[List[int]], List[int]]:
181     total_time = sum(times)
182     max_time = max(times)
183
184     lb = max(max_time, math.ceil(total_time / m))
185     ub = total_time
186
187     best_c = None
188     best_stations = None
189     best_loads = None
190
191     while lb <= ub:

```

```

192         mid = (lb + ub) // 2
193         feasible, stations, loads = decode_with_cycle_limit(
194             perm, times, preds, succs, m, mid
195         )
196         if feasible:
197             best_c = mid
198             best_stations = stations
199             best_loads = loads
200             ub = mid - 1
201         else:
202             lb = mid + 1
203
204     if best_c is None:
205         return float("inf"), [[] for _ in range(m)], [0 for _ in
206             range(m)]
207
208     return float(best_c), best_stations, best_loads
209
210 # =====
211 # 5) GA OPERATORLERI
212 # =====
213
214 def tournament_selection(
215     population: List[List[int]],
216     fitness: List[float],
217     rng: random.Random,
218     k: int = 2
219 ) -> List[int]:
220     n = len(population)
221     best_idx = None
222     best_fit = -float("inf")
223     for _ in range(k):
224         i = rng.randrange(n)
225         if fitness[i] > best_fit:
226             best_fit = fitness[i]
227             best_idx = i
228     return population[best_idx][:]
229
230
231 def pox_crossover(

```

```

232     p1: List[int],
233     p2: List[int],
234     rng: random.Random
235 ) -> List[int]:
236     """
237     POX crossover + istege bagli debug ciktilari.
238     """
239     global pox_debug_counter
240
241     chosen = set()
242     for gene in p1:
243         if rng.random() < 0.5:
244             chosen.add(gene)
245     if not chosen:
246         chosen.add(rng.choice(p1))
247
248     child = []
249     for gene in p1:
250         if gene in chosen:
251             child.append(gene)
252     for gene in p2:
253         if gene not in chosen:
254             child.append(gene)
255
256     # ---- Debug output ----
257     if DEBUG_POX and pox_debug_counter < POX_DEBUG_LIMIT:
258         pox_debug_counter += 1
259         print("\n== POX CROSSOVER SAMPLE ==")
260         print("Parent1 (first 15):", p1[:15])
261         print("Parent2 (first 15):", p2[:15])
262         print("Chosen subset:", sorted(chosen))
263         print("Child (first 15):", child[:15])
264         print("=====\n")
265
266     return child
267
268
269 def swap_mutation(perm: List[int], rng: random.Random) -> List[int]
270 ]:
271     n = len(perm)
272     i, j = rng.sample(range(n), 2)

```

```

272     perm[i], perm[j] = perm[j], perm[i]
273     return perm
274
275
276 # =====
277 # 6) GA: TEK RUN + ISTATISTIK
278 # =====
279
280 def genetic_algorithm_ualbp2(
281     n: int,
282     times: List[int],
283     preds: Dict[int, Set[int]],
284     succs: Dict[int, Set[int]],
285     m: int,
286     pop_size: int,
287     generations: int,
288     crossover_rate: float,
289     mutation_rate: float,
290     seed: int,
291     run_index: int
292 ):
293     rng = random.Random(seed)
294
295     population: List[List[int]] = []
296     for _ in range(pop_size):
297         perm = random_topological_sort(n, preds, rng)
298         population.append(perm)
299
300     best_perm = None
301     best_cycle = float("inf")
302     best_stations = None
303     best_loads = None
304
305     # ----- CONVERGENCE KAYDI (jenerasyon bazli) -----
306     history_best_so_far = []
307     history_best_gen = []
308     history_mean_gen = []
309
310     for gen in range(generations):
311         fitness_vals = []
312         eval_cache = {}

```

```

313
314     for perm in population:
315         key = tuple(perm)
316         if key in eval_cache:
317             cycle_time = eval_cache[key]
318         else:
319             cycle_time, stations, loads =
320                 evaluate_permutation_cycle_time(
321                     perm, times, preds, succs, m
322                 )
323             eval_cache[key] = cycle_time
324             if cycle_time < best_cycle:
325                 best_cycle = cycle_time
326                 best_perm = perm[:]
327                 best_stations = stations
328                 best_loads = loads
329
330             if math.isinf(eval_cache[key]):
331                 fit = 0.0
332             else:
333                 fit = 1.0 / eval_cache[key]
334             fitness_vals.append(fit)
335
336     gen_cycles = list(eval_cache.values())
337     gen_best = min(gen_cycles)
338     gen_mean = sum(gen_cycles) / len(gen_cycles)
339
340     history_best_gen.append(gen_best)
341     history_mean_gen.append(gen_mean)
342     history_best_so_far.append(best_cycle)
343
344     new_population: List[List[int]] = []
345     if best_perm is not None:
346         new_population.append(best_perm[:])
347
348     while len(new_population) < pop_size:
349         p1 = tournament_selection(population, fitness_vals, rng
350             , k=2)

```

```

351         if rng.random() < crossover_rate:
352             child = pox_crossover(p1, p2, rng)
353         else:
354             child = p1[:]
355
356         if rng.random() < mutation_rate:
357             child = swap_mutation(child, rng)
358
359         child = repair_to_topological(child, preds)
360         new_population.append(child)
361
362     population = new_population
363
364     # ----- RUN SONU: final populasyon istatistikleri -----
365     final_stats = []
366     cycles = []
367     fits = []
368
369     for perm in population:
370         cycle_time, _, _ = evaluate_permutation_cycle_time(
371             perm, times, preds, succs, m
372         )
373         fit = 0.0 if math.isinf(cycle_time) else 1.0 / cycle_time
374         cycles.append(cycle_time)
375         fits.append(fit)
376         final_stats.append((fit, cycle_time, perm))
377
378     best_cycle_final = min(cycles)
379     worst_cycle_final = max(cycles)
380     mean_cycle_final = sum(cycles) / len(cycles)
381
382     best_fit_final = max(fits)
383     worst_fit_final = min(fits)
384     mean_fit_final = sum(fits) / len(fits)
385
386     final_stats.sort(key=lambda x: x[0], reverse=True)
387     top10 = final_stats[:10]
388
389     run_stats = {
390         "best_cycle": best_cycle,
391         "final_best_cycle": best_cycle_final,

```

```

392     "final_mean_cycle": mean_cycle_final,
393     "final_worst_cycle": worst_cycle_final,
394     "final_best_fit": best_fit_final,
395     "final_mean_fit": mean_fit_final,
396     "final_worst_fit": worst_fit_final,
397     "top10": top10,
398     "history_best_so_far": history_best_so_far,
399     "history_best_gen": history_best_gen,
400     "history_mean_gen": history_mean_gen,
401 }
402
403 return best_perm, best_cycle, best_stations, best_loads,
404        run_stats
405
406 # =====
407 # 7) COKLU RUN + OZET
408 # =====
409
410 def main():
411     start_time = time.time()
412
413     n, times, precedences = read_in2(INSTANCE_PATH)
414     preds, succs = build_pred_succ(n, precedences)
415
416     total_time = sum(times)
417     max_time = max(times)
418     lb1 = math.ceil(total_time / M_STATIONS)
419     lb2 = max_time
420     lb = max(lb1, lb2)
421
422     print(f"Instance: {INSTANCE_PATH}")
423     print(f"Tasks (n)           : {n}")
424     print(f"Stations (m)           : {M_STATIONS}")
425     print(f"Total work content sum : {total_time}")
426     print(f"Max task time          : {max_time}")
427     print(f"Lower bound LB1=ceil(sum/m) = {lb1}")
428     print(f"Lower bound LB2=max(t_i)   = {lb2}")
429     print(f"Overall theoretical LB     = {lb}")
430     print()
431

```

```

432 all_convergence_rows = []
433 run_best_cycles = []
434 run_runtimes = []
435 best_overall_cycle = float("inf")
436 best_overall_solution = None
437 best_overall_top10 = None
438
439 for r in range(1, RUNS + 1):
440     seed = BASE_SEED + r
441     print(f"\n=====")
442     print(f"          RUN {r}/{RUNS}")
443     print(f"=====")
444     run_start = time.time()
445
446     best_perm, best_cycle, best_stations, best_loads, stats =
         genetic_algorithm_ualbp2(
447         n=n,
448         times=times,
449         preds=preds,
450         succs=succs,
451         m=M_STATIONS,
452         pop_size=POP_SIZE,
453         generations=GENERATIONS,
454         crossover_rate=CROSSOVER_RATE,
455         mutation_rate=MUTATION_RATE,
456         seed=seed,
457         run_index=r
458     )
459
460     hb = stats["history_best_so_far"]
461     hbg = stats["history_best_gen"]
462     hm = stats["history_mean_gen"]
463
464     for g in range(GENERATIONS):
465         all_convergence_rows.append([r, g + 1, hb[g], hbg[g],
466                                     hm[g]])
467
468     run_end = time.time()
469     run_runtime = run_end - run_start
470
471     run_best_cycles.append(best_cycle)

```



```

471         run_runtimes.append(run_runtime)
472
473     print(f"\nRun {r} summary:")
474     print(f"    Best cycle time over generations      : {
        best_cycle}")
475     print(f"    Final population best / mean / worst c: "
476           f"{stats['final_best_cycle']} / {stats['
        final_mean_cycle']:.2f} / {stats['
        final_worst_cycle']}")
477     print(f"    Final population best / mean / worst fitness: "
478           f"{stats['final_best_fit']:.6f} / {stats['
        final_mean_fit']:.6f} / {stats['final_worst_fit
        ']:.6f}")
479
480     print("\n    Top 10 individuals in final population:")
481     for rank, (fit, cyc, perm) in enumerate(stats["top10"],
        start=1):
482         print(f"        {rank:2d}) fitness={fit:.6f}, cycle={cyc}")
483         feasible, stations, loads = decode_with_cycle_limit(
484             perm, times, preds, succs, M_STATIONS, int(cyc)
485         )
486         for s_idx, tasks in enumerate(stations, start=1):
487             print(f"            Station {s_idx}: {tasks}")
488
489     if best_cycle < best_overall_cycle:
490         best_overall_cycle = best_cycle
491         best_overall_solution = (best_perm, best_stations,
            best_loads)
492         best_overall_top10 = stats["top10"]
493
494     mean_cycle_runs = sum(run_best_cycles) / len(run_best_cycles)
495     worst_cycle_runs = max(run_best_cycles)
496
497     best_gap_abs = best_overall_cycle - OPTIMAL_CYCLE
498     best_gap_pct = 100.0 * best_gap_abs / OPTIMAL_CYCLE
499
500     mean_gap_abs = mean_cycle_runs - OPTIMAL_CYCLE
501     mean_gap_pct = 100.0 * mean_gap_abs / OPTIMAL_CYCLE
502
503     total_runtime = time.time() - start_time
504

```

```

505     print("\n=====")
506     print("          10 RUN GLOBAL SUMMARY")
507     print("=====")
508     print(f"GA parameters:")
509     print(f"  Population size      : {POP_SIZE}")
510     print(f"  Generations           : {GENERATIONS}")
511     print(f"  Crossover rate        : {CROSSOVER_RATE}")
512     print(f"  Mutation rate         : {MUTATION_RATE}")
513     print(f"  Parent selection      : Tournament (k=2)")
514     print(f"  Crossover method      : POX-benzeri precedence preserving
515           ")
516     print(f"  Mutation method       : Swap mutation")
517     print(f"  Iteration number      : {GENERATIONS} generations per run
518           ")
519     print(f"  Total runs            : {RUNS}")
520     print()
521     print(f"Run best cycle times: {run_best_cycles}")
522     print(f"Best  cycle over runs : {best_overall_cycle}")
523     print(f"Mean  cycle over runs : {mean_cycle_runs:.2f}")
524     print(f"Worst cycle over runs : {worst_cycle_runs}")
525     print()
526     print(f"Known optimal C*       : {OPTIMAL_CYCLE}")
527     print(f"Best-of-10 gap         : {best_gap_abs} ({best_gap_pct
528           :.3f} % above optimum)")
529     print(f"Mean-of-10 gap         : {mean_gap_abs:.2f} ({
530           mean_gap_pct:.3f} % above optimum)")
531     print()
532     best_perm, best_stations, best_loads = best_overall_solution
533     print("Best-of-10 solution station loads:")
534     for i, load in enumerate(best_loads, start=1):
535         print(f"  Station {i}: load = {load}")
536     print("\nBest-of-10 solution assignments:")
537     for i, tasks in enumerate(best_stations, start=1):
538         print(f"  Station {i}: {tasks}")
539     print("\nTop 10 individuals of best run (by fitness):")

```

```

541     for rank, (fit, cyc, perm) in enumerate(best_overall_top10,
        start=1):
542         print(f"    {rank:2d}) fitness={fit:.6f}, cycle={cyc}")
543         feasible, stations, loads = decode_with_cycle_limit(
544             perm, times, preds, succs, M_STATIONS, int(cyc)
545         )
546         for s_idx, tasks in enumerate(stations, start=1):
547             print(f"            Station {s_idx}: {tasks}")
548
549     print(f"\nTotal runtime for {RUNS} runs: {total_runtime:.2f}
        seconds")
550
551     instance_name = os.path.splitext(os.path.basename(INSTANCE_PATH
        ))[0]
552     timestamp = datetime.now().strftime("%Y-%m-%d_%H-%M")
553
554     summary_file      = f"{instance_name}_summary_{timestamp}.csv"
555     runs_file         = f"{instance_name}_run_details_{timestamp}.
        csv"
556     best_details_file = f"{instance_name}_best_details_{timestamp}.
        csv"
557     top10_file        = f"{instance_name}-top-10-individuals-{
        timestamp}.csv"
558     convergence_file  = f"{instance_name}_convergence_{timestamp}.
        csv"
559
560     print("\nGenerated output filenames:")
561     print("    Summary file      :", summary_file)
562     print("    Run details file   :", runs_file)
563     print("    Best details file  :", best_details_file)
564     print("    Top10 file         :", top10_file)
565     print()
566
567     with open(summary_file, "w", newline="", encoding="utf-8") as f
        :
568         writer = csv.writer(f)
569         writer.writerow([
570             "Instance", "Stations(m)", "TotalWork(sum_ti)", "
                Population", "Generations",
571             "CrossoverRate", "MutationRate", "ParentSelection", "
                CrossoverMethod",

```

```

572         "MutationMethod", "Runs", "BestCycle", "MeanCycle", "
           WorstCycle", "OptimalC",
573         "BestGapAbs", "BestGapPct", "MeanGapAbs", "MeanGapPct", "
           TotalRuntime(sec)"
574     ])
575     writer.writerow([
576         instance_name, M_STATIONS, total_time, POP_SIZE,
           GENERATIONS,
577         CROSSOVER_RATE, MUTATION_RATE, "Tournament(k=2)", "POX"
           , "Swap", RUNS,
578         best_overall_cycle, f"{mean_cycle_runs:.2f}",
           worst_cycle_runs, OPTIMAL_CYCLE,
579         best_gap_abs, f"{best_gap_pct:.3f}", f"{mean_gap_abs:.2
           f}", f"{mean_gap_pct:.3f}",
580         f"{total_runtime:.2f}"
581     ])
582
583     print(f"CSV summary saved to: {summary_file}\n")
584
585     with open(runs_file, "w", newline="", encoding="utf-8") as f:
586         writer = csv.writer(f)
587         writer.writerow(["Run", "BestCycle", "RunRuntime(sec)"])
588         for r in range(len(run_best_cycles)):
589             writer.writerow([r + 1, run_best_cycles[r], f"{
                 run_runtimes[r]:.2f}"])
590
591     print(f"Run details saved to: {runs_file}")
592
593     total_line_eff = 100.0 * total_time / (M_STATIONS *
        best_overall_cycle)
594
595     with open(best_details_file, "w", newline="", encoding="utf-8")
        as f:
596         writer = csv.writer(f)
597         writer.writerow([
598             "Instance", "Stations(m)", "BestCycle", "TotalWork(sum_ti)
                ", "GlobalLineEfficiency(%)",
599             "StationIndex", "StationLoad", "StationUtilization(Load/
                BestCycle)", "TasksSequence"
600         ])

```

```

601     for s_idx, (load, tasks) in enumerate(zip(best_loads,
        best_stations), start=1):
602         utilization = load / best_overall_cycle
603         tasks_str = " ".join(str(t) for t in tasks)
604         writer.writerow([
605             instance_name, M_STATIONS, best_overall_cycle,
606             total_time,
607             f"{total_line_eff:.2f}", s_idx, load, f"{
        utilization:.4f}", tasks_str
608     ])
609
610     print(f"Best solution details saved to: {best_details_file}")
611
612     with open(top10_file, "w", newline="", encoding="utf-8") as f:
613         writer = csv.writer(f)
614         writer.writerow([
615             "Instance", "Stations(m)", "RankInBestRun", "Fitness", "
        Cycle", "StationIndex",
616             "StationLoad", "StationUtilization(Load/Cycle)", "
        TasksSequence"
617         ])
618
619     for rank, (fit, cyc, perm) in enumerate(best_overall_top10,
        start=1):
620         feasible, stations, loads = decode_with_cycle_limit(
621             perm, times, preds, succs, M_STATIONS, int(cyc)
622         )
623         for s_idx, (load, tasks) in enumerate(zip.loads,
        stations), start=1):
624             tasks_str = " ".join(str(t) for t in tasks)
625             utilization = 0.0 if cyc == 0 else load / cyc
626             writer.writerow([
627                 instance_name, M_STATIONS, rank, f"{fit:.6f}",
628                 cyc,
629                 s_idx, load, f"{utilization:.4f}", tasks_str
630             ])
631
632     print(f"Top 10 individuals (best run) saved to: {top10_file}")
633
634     with open(convergence_file, "w", newline="", encoding="utf-8")
        as f:
635         writer = csv.writer(f)

```

```
633         writer.writerow(["Run", "Generation", "BestSoFar", "
        BestInGeneration", "MeanCycle"])
634     writer.writerows(all_convergence_rows)
635
636     print(f"Convergence history saved to: {convergence_file}")
637
638
639 if __name__ == "__main__":
640     main()
```

Table 3: Run-by-run best cycle time and runtime results for all benchmark instances.

(a) BARTHOLD

Run	BestCycle	Runtime (s)
1	470	40.56
2	471	39.63
3	471	38.53
4	471	41.16
5	471	43.95
6	471	43.33
7	471	40.14
8	471	43.86
9	471	44.41
10	471	43.95

(b) TONGE70

Run	BestCycle	Runtime (s)
1	210	22.10
2	210	25.35
3	209	22.37
4	210	26.65
5	210	22.35
6	209	20.31
7	210	21.34
8	210	20.61
9	210	20.40
10	210	20.53

(c) SAWYER30

Run	BestCycle	Runtime (s)
1	47	3.00
2	47	3.37
3	47	3.15
4	47	3.29
5	47	3.22
6	47	3.32
7	47	3.54
8	47	3.24
9	47	3.69
10	47	3.61

(d) LUTZ1

Run	BestCycle	Runtime (s)
1	1400	7.93
2	1400	8.13
3	1400	7.89
4	1400	8.04
5	1400	8.05
6	1400	7.92
7	1400	7.83
8	1400	7.53
9	1400	7.64
10	1400	7.52

Table 3: Run-by-run best cycle time and runtime results for all benchmark instances (continued).

(e) ARC111			(f) LUTZ3		
Run	BestCycle	Runtime (s)	Run	BestCycle	Runtime (s)
1	12542	44.32	1	548	9.68
2	12546	43.13	2	548	8.93
3	12549	43.63	3	548	8.30
4	12547	43.07	4	548	8.27
5	12549	42.76	5	548	8.23
6	12546	43.46	6	548	8.72
7	12550	40.49	7	548	8.60
8	12549	42.95	8	548	8.15
9	12550	42.20	9	548	8.54
10	12543	42.57	10	548	8.37

(g) SCHOLL			(h) ARC83		
Run	BestCycle	Runtime (s)	Run	BestCycle	Runtime (s)
1	1825	275.45	1	6467	25.89
2	1827	290.46	2	6562	26.65
3	1821	299.74	3	6488	25.94
4	1824	285.49	4	6534	25.53
5	1818	273.51	5	6562	25.87
6	1829	291.93	6	6508	25.50
7	1821	284.44	7	6512	26.62
8	1827	285.68	8	6560	30.12
9	1823	287.15	9	6544	26.70
10	1823	334.99	10	6467	29.45

B Run-by-run Results for All Benchmark Instances

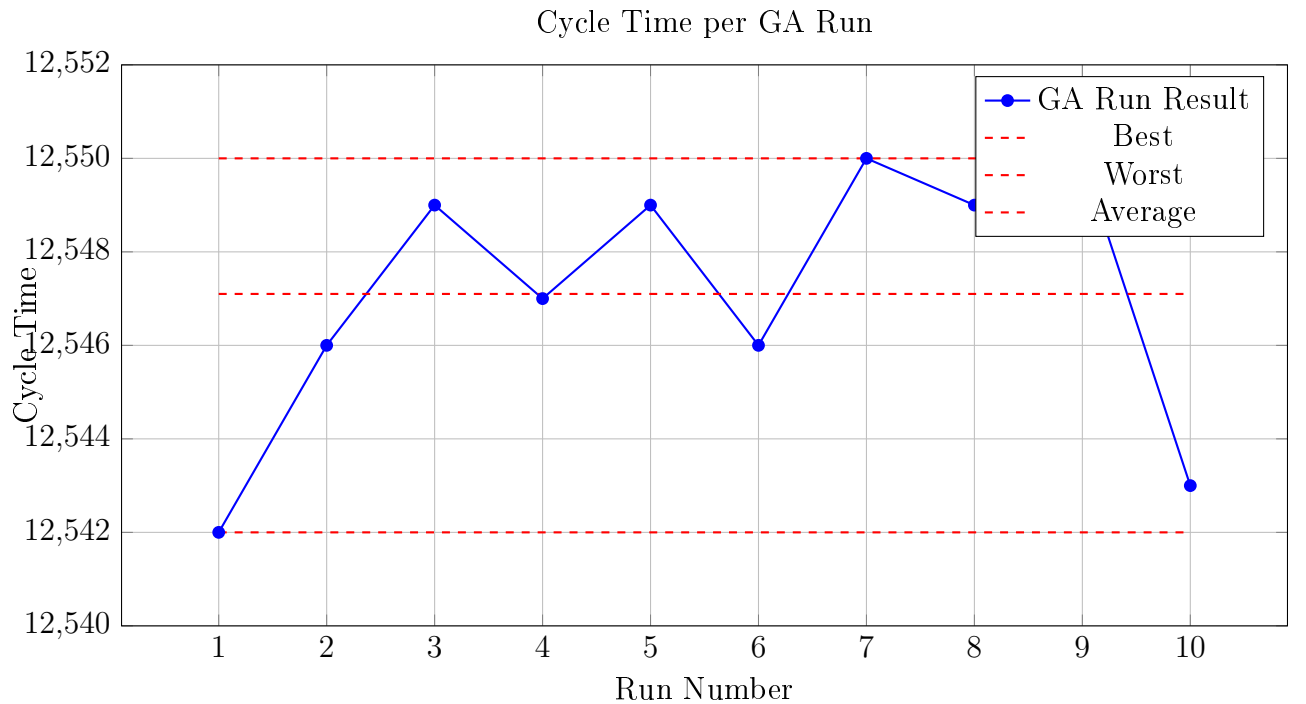


Figure 10: Instance ARC83, Cycle time obtained in each of the 10 independent GA runs, including best, worst, and average values. ⁴⁷

Table 3: Run-by-run best cycle time and runtime results for all benchmark instances (continued).

(i) BARTHOL2			(j) GUNTHER		
Run	BestCycle	Runtime (s)	Run	BestCycle	Runtime (s)
1	159	66.58	1	44	5.09
2	159	71.32	2	44	5.46
3	159	69.39	3	44	6.34
4	159	61.17	4	44	6.25
5	159	60.83	5	44	5.30
6	159	60.94	6	44	5.04
7	159	61.24	7	44	5.49
8	159	62.09	8	44	6.48
9	159	63.06	9	44	5.58
10	159	61.65	10	44	5.44

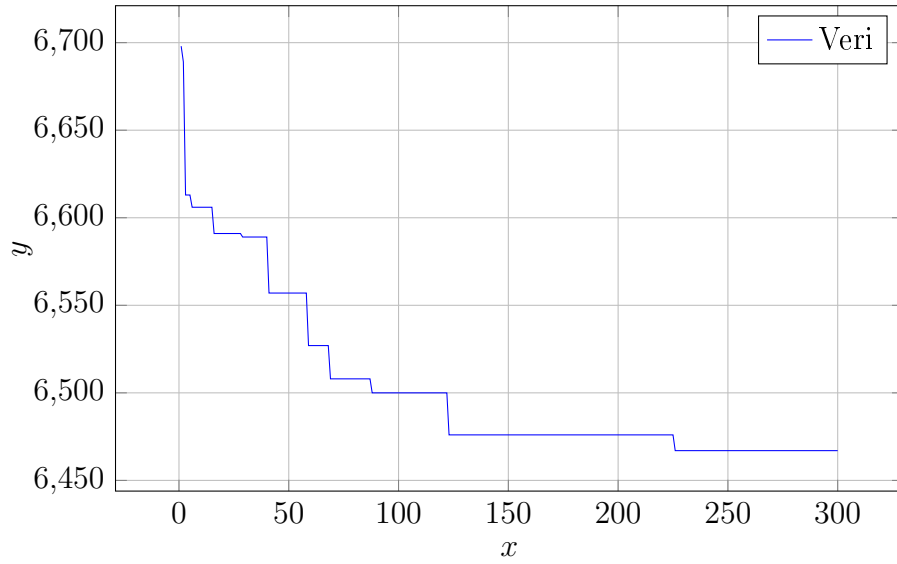


Figure 11: CSV dosyasından çizilen grafik

C Additional Results for ARC83' top 10 Individuals

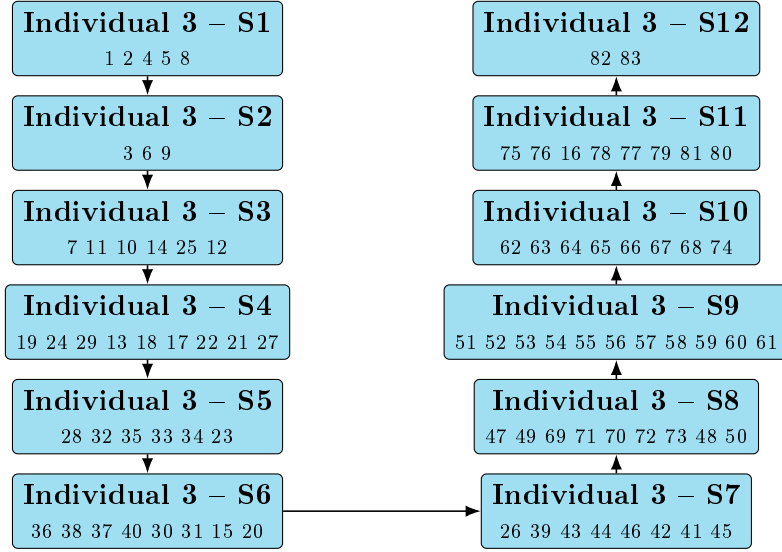


Figure 12: U-shaped 12-station assignment for ARC83 (Individual 3, Fitness = 0.000152).

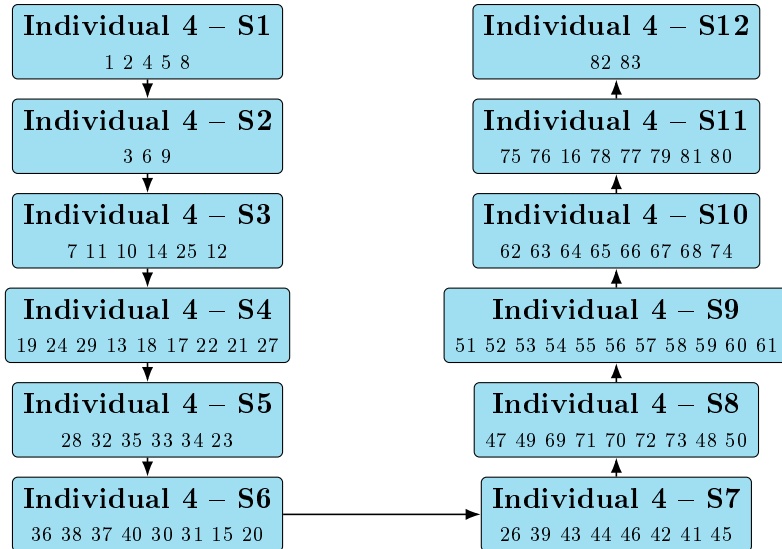


Figure 13: U-shaped 12-station assignment for ARC83 (Individual 4, Fitness = 0.000152).

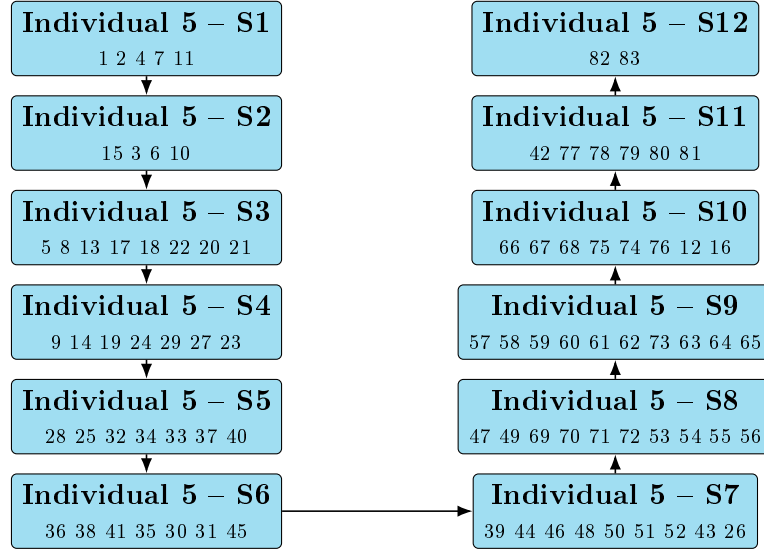


Figure 14: U-shaped 12-station assignment for ARC83 (Individual 5, Fitness = 0.000151).

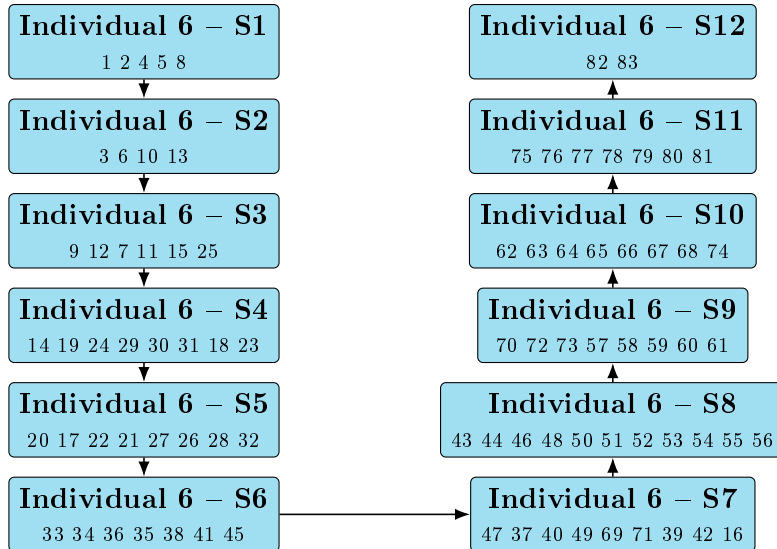


Figure 15: U-shaped 12-station assignment for ARC83 (Individual 6, Fitness = 0.000151).

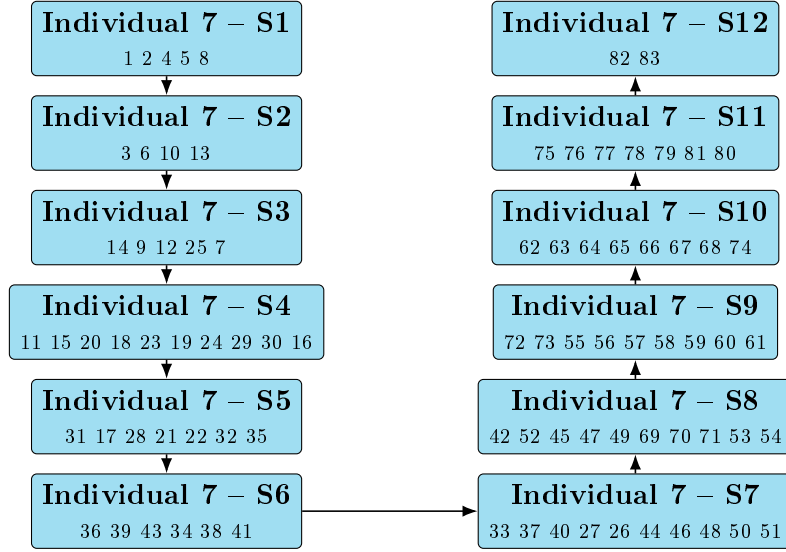


Figure 16: U-shaped 12-station assignment for ARC83 (Individual 7, Fitness = 0.000151).

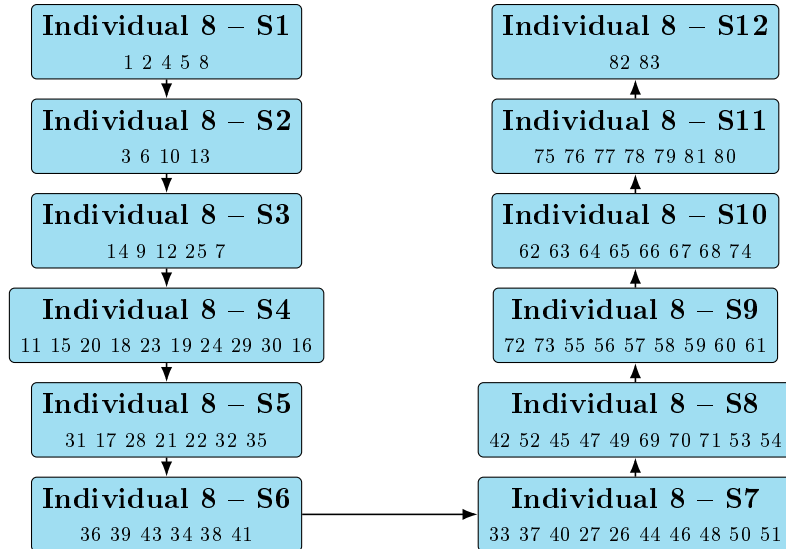


Figure 17: U-shaped 12-station assignment for ARC83 (Individual 8, Fitness = 0.000151).

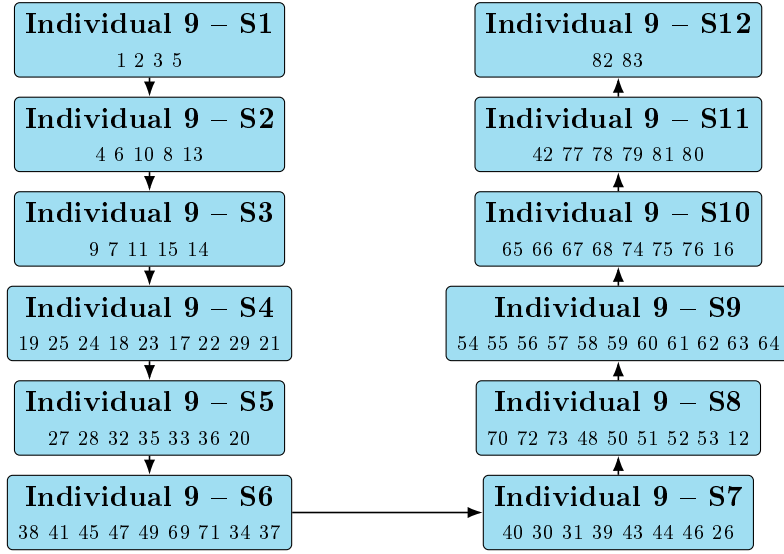


Figure 18: U-shaped 12-station assignment for ARC83 (Individual 9, Fitness = 0.000151).

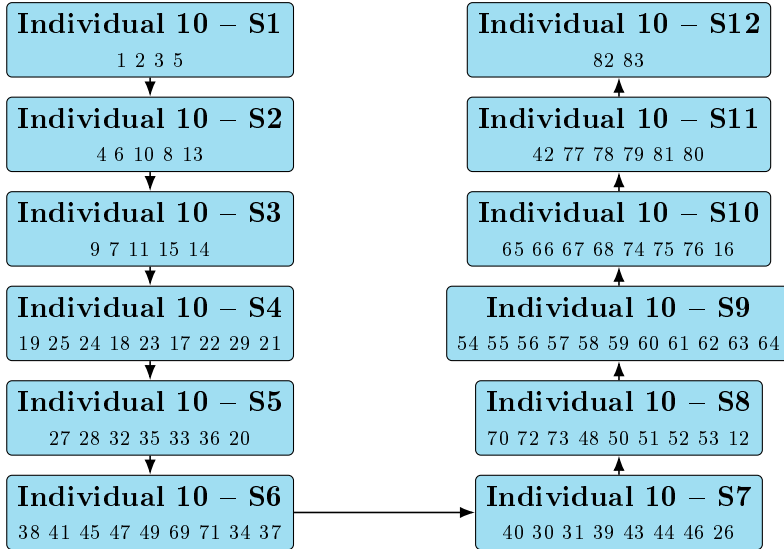


Figure 19: U-shaped 12-station assignment for ARC83 (Individual 10, Fitness = 0.000151).